GNN-SDM: Species Distribution Modelling with Graph Neural Networks for Swiss Flora at 30 m Resolution

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Abstract

Traditional species distribution models treat occurrences as independent points, ignoring the spatial structure of landscapes that governs habitat connectivity and species dispersal. We adapt the GNN-SDM framework (Wu et al., 2025) to Swiss vascular plants at 30 m resolution, constructing a landscape graph of 166,464 patches linked by 465,833 adjacency edges. Using a GraphSAGE architecture tuned for flora, we model habitat suitability for 3,751 species and compare against a Random Forest baseline. [Results summary to be added.]

Contents

1 Introduction

Biodiversity is declining at an unprecedented rate worldwide, driven by habitat loss, fragmentation, climate change, and land-use intensification (IPBES, 2019). Switzerland, despite its relatively small area, harbours exceptional plant diversity owing to its pronounced altitudinal gradients and mosaic of biogeographic regions. Yet even here, a third of all assessed species and half of all habitat types are threatened (Swiss Federal Council, 2022), with 28% of vascular plants considered endangered or regionally extinct and an additional 14% near-threatened (**InfoFlora2024**). The main drivers—soil sealing, landscape fragmentation, intensive land use, and nitrogen inputs—continue to erode habitat quality and connectivity (Swiss Federal Council, 2022; Federal Office for the Environment, 2025). Effective conservation planning requires spatially explicit predictions of where species can persist, and crucially, which landscape elements link suitable habitat patches into functional networks.

Species distribution models (SDMs) have become a cornerstone of conservation biogeography, relating georeferenced occurrence records to environmental predictors to map habitat suitability across space (Elith and Leathwick, 2009; Guisan and Thuiller, 2005). Classical approaches, such as generalised linear models, maximum entropy (Phillips et al., 2006), and ensemble methods such as Random Forests (Breiman, 2001), treat each occurrence as an independent sample drawn from an environmental niche. While powerful, this point-based paradigm ignores the spatial structure of landscapes: two patches with identical environmental conditions may differ dramatically in ecological value depending on their size, shape, and connectivity to neighbouring habitat. For species whose long-term persistence depends on seed dispersal between habitat patches, such as the Cypripedium calceolus, this omission can lead to misleading suitability estimates and suboptimal conservation recommendations.

Graph Neural Networks (GNNs) offer a principled way to incorporate landscape structure into SDMs. By representing the landscape as a graph, where nodes are ecologically homogeneous patches and edges encode spatial adjacency. A GNN can learn how environmental conditions in neighbouring patches jointly influence local habitat suitability. Wu et al. (2025) introduced the SOM-GNN-SDM framework, demonstrating that a GraphSAGE architecture (Hamilton et al., 2017) trained on landscape graphs substantially outperforms traditional SDMs for virtual and real species across continental extents. Their approach constructs landscape patches via Self-Organising Maps (SOMs) (Kohonen, 1982), builds an adjacency graph, and propagates information across the graph to produce suitability predictions that are spatially coherent and connectivity-aware.

Beyond improving predictive accuracy, the graph representation opens a direct path to conservation applications. The Swiss federal government has identified the lack of an ecological infrastructure that protects and connects core biodiversity areas as a key deficit, setting a target of 17% of national territory as connected core areas by 2030, up from 13.4% today (Swiss Federal Council, 2022; Federal Office for the Environment, 2025). Because the landscape graph explicitly encodes connectivity, it can be analysed using circuit theory and centrality metrics to identify critical corridors: patches whose protection would maintain or restore habitat connectivity for threatened species. This transforms the SDM output from a static suitability map into an actionable tool for spatial conservation prioritisation and gap analysis, directly informing where the ecological infrastructure should be expanded.

In this work, we adapt the GNN-SDM framework to Swiss vascular flora at 30 m resolution, presenting a substantially finer grain than the original continental-scale application. We construct a landscape graph of 166,464 patches from 12 environmental features using SOM-based clustering and connected-component labelling, train per-species GraphSAGE models for 3,751 qualifying plant species, and benchmark against a Random Forest baseline. We further apply circuit-theory connectivity analysis to the resulting suitability maps to identify conservation priority areas and quantify protection gaps relative to the existing protected-area network.

Specifically, we address the following research questions:

1. Can GNN-SDM outperform a traditional Random Forest at patch-level species distribution modelling for Swiss plants?
2. Does the graph structure confer greater benefit for rare and vulnerable species, whose distributions are more fragmented?
3. What GNN architecture is best suited for flora—given that plants, unlike the mobile fauna studied by Wu et al. (2025), have limited active dispersal?
4. Can the graph structure be leveraged to identify spatial gaps in the protected-area network where additional conservation measures would be most beneficial?

2 Data and Study Area

2.1 Study Area

The study area encompasses the entirety of Switzerland and a surrounding buffer zone, defined by the spatial extent of the CHELSAch high-resolution climate dataset (WSL, 2024). The bounding box spans from 45.57° N to 48.07° N latitude and 5.72° E to 10.66° E longitude, covering approximately 41,400 km^2 of land area. This extent includes a border buffer of roughly 10 km into neighbouring countries to avoid edge effects in spatial analyses such as flow accumulation and neighbourhood-based feature computation.

All environmental layers are aligned to a common 30 m master grid derived from the Copernicus GLO-30 Digital Elevation Model (European Space Agency, 2023), projected in geographic coordinates (EPSG:4326). Switzerland's topography ranges from 193 m (Lake Maggiore) to 4,634 m (Dufourspitze), creating steep environmental gradients over short distances—a property that motivates the use of high-resolution modelling.

2.2 Environmental Features

We compiled 21 candidate environmental features from four thematic groups (Table 1): terrain, land cover, climate, and biological/anthropogenic variables. All layers were processed to the 30 m master grid using source-appropriate resampling methods.

Terrain features. Elevation was obtained directly from the Copernicus GLO-30 DEM (European Space Agency, 2023). Slope and aspect were derived using first-order finite differences at 30 m resolution. Aspect was encoded as two variables (sin and cos of the azimuth angle) to avoid the circular discontinuity at 0°/360°. The Topographic Wetness Index (TWI) was computed using D8 flow direction and accumulation via the `pysheds` library, with pit-filling and depression resolution applied beforehand. Profile and plan curvature were derived from second-order finite differences and smoothed with a 5×5 mean filter to reduce noise.

Land cover. Six land-cover fraction layers were computed from ESA WorldCover 2021 (European Space Agency, 2022) at its native 10 m resolution. For each class, a binary mask was reprojected to the 30 m grid using average resampling, yielding the fraction of each 30 m pixel covered by that land-cover type. The six classes retained for Switzerland are: tree cover, grassland, cropland, built-up, snow/ice, and permanent water bodies.

Climate. Five bioclimatic variables were derived from CHELSAch monthly temperature and precipitation normals (1981–2010) at 25 m resolution (WSL, 2024): annual mean temperature (Bio1), temperature seasonality (Bio4, standard deviation), annual precipitation (Bio12), precipitation seasonality (Bio15, coefficient of variation), and precipitation of the warmest quarter (Bio18). Each variable was regridded to the 30 m master grid using bilinear interpolation.

Table 1: Environmental predictor variables compiled for the study. All layers are aligned to the 30 m master grid. The 12 features selected for modelling (after feature selection, Section 2.5) are marked with •.

Group	Feature	Source	Native res.	Sel.
Terrain	Elevation	Copernicus GLO-30 DEM	30 m	•
	Slope	Derived from DEM	30 m	•
	Aspect (sin, cos)	Derived from DEM	30 m	•
	Topographic Wetness Index	D8 flow accumulation	30 m	•
	Profile curvature	2^{nd}-order finite diff.	30 m	
	Plan curvature	2^{nd}-order finite diff.	30 m	
Land cover	Tree cover fraction	ESA WorldCover 2021	10 m	•
	Grassland fraction	ESA WorldCover 2021	10 m	•
	Cropland fraction	ESA WorldCover 2021	10 m	•
	Built-up fraction	ESA WorldCover 2021	10 m	
	Snow/ice fraction	ESA WorldCover 2021	10 m	
	Water fraction	ESA WorldCover 2021	10 m	
Climate	Bio1 (annual mean temp.)	CHELSAch	25 m	•
	Bio4 (temp. seasonality)	CHELSAch	25 m	•
	Bio12 (annual precip.)	CHELSAch	25 m	•
	Bio15 (precip. seasonality)	CHELSAch	25 m	
	Bio18 (precip. warmest quarter)	CHELSAch	25 m	
Biological	Canopy height	ETH Global Canopy Height	10 m	•
	NDVI (annual max)	GIMMS MODIS	250 m	
	Human Footprint	Mu et al. (2022)	1 km	•
	Forest-edge distance	Derived from tree cover	30 m	

Biological and anthropogenic features. Canopy height was obtained from the ETH Global Canopy Height Model at 10 m resolution (Lang et al., 2023) and regridded to 30 m via bilinear resampling. The annual maximum NDVI was computed from GIMMS MODIS 8-day composites (250 m, growing season DOY 97–273) and regridded to 30 m. The global Human Footprint index (Mu et al., 2022) at 1 km was bilinearly downscaled. Forest-edge distance was computed as a signed Euclidean distance transform from the tree-cover fraction layer (threshold 0.5): positive values indicate distance from forest, negative values indicate depth within forest.

2.3 Species Occurrence Data

Species occurrence records for vascular plants in Switzerland were obtained from the Global Biodiversity Information Facility (GBIF) open data snapshot hosted on Amazon S3 (GBIF.org, 2024). The raw dataset was filtered to retain only records within the study extent that met the following quality criteria: coordinate uncertainty ≤ 1000 m, valid geographic coordinates, and taxonomic rank at species level.

After filtering, the dataset comprised approximately 14 million occurrence records spanning 7,912 vascular plant species. For modelling purposes, we retained only species with at least 100 unique occurrence records, yielding 3,756 qualifying species. This threshold ensures sufficient training data for both the Random Forest baseline and the GNN-SDM, while acknowledging that it excludes many rare and threatened species with sparse observational records.

2.4 Protected Areas

Protected area boundaries for Switzerland were obtained from the World Database on Protected Areas (WDPA), filtered to nationally designated sites within the study extent. The dataset includes national parks, nature reserves, landscape protection areas, and Ramsar sites. These boundaries are used in the gap analysis (??) to assess the degree to which high-priority habitat identified by the GNN-SDM falls within existing protected areas.

2.5 Feature Selection

To reduce multicollinearity and computational cost, we performed feature selection using Random Forest importance scores. A patch-level RF classifier was trained for six representative species spanning different habitat types and prevalence levels. Feature importance was computed as the mean decrease in Gini impurity, averaged across species.

From the initial 21 candidate features, 12 were retained for downstream modelling (Table 1). The excluded features were either highly correlated with retained variables (e.g., Bio15 with Bio12; plan curvature with profile curvature) or contributed negligible predictive power (e.g., NDVI at 250 m resolution added little beyond the 10 m canopy height and tree-cover fraction). The selected feature set balances ecological interpretability with predictive performance while keeping the dimensionality manageable for the Self-Organising Map clustering step.

3 Methods

This section describes the methodological pipeline for constructing a graph-based species distribution model (SOM-GNN-SDM) for Swiss flora at 30 m resolution, following the framework proposed by Wu et al. (2025). The pipeline consists of three stages: landscape patch construction via Self-Organising Map clustering and connected- component analysis, (2) a Random Forest baseline, and (3) the SOM-GNN-SDM itself.

3.1 Landscape Patch Construction

The landscape patch construction transforms the continuous 30 m raster feature stack into a graph representation where nodes correspond to ecologically homogeneous habitat patches and edges encode spatial adjacency.

3.1.1 SOM Clustering

A Self-Organising Map (SOM; Kohonen, 1982) was trained to identify recurring landscape types from the 12 selected environmental features. The SOM was configured as a 64×64 hexagonal, toroidal grid and initialised with PCA. Training used a subsample of 500 000 pixels (drawn from the ~160 M valid pixels in the study extent), preprocessed with a `RobustScaler` and domain-specific post-scaling (land-cover fractions up-weighted 3×; forest-edge distance clipped to $[-5, 5]$).

The SOM was trained for 150 epochs with a neighbourhood radius decaying from 64 to 16 and a learning rate decaying from 0.05 to 0.016. Convergence was assessed on a held-out 20% validation set using quantisation error (QE) and topographic error (TE); 150 epochs represented the sweet spot beyond which improvements were marginal.

After training, the codebook vectors were hierarchically clustered using Ward's linkage, and the dendrogram was cut to yield 30 landscape types. Every valid pixel in the study extent was then assigned to its best-matching unit (BMU) via a FAISS exact-L_2 index, producing a categorical landscape-type raster at 30 m resolution.

3.1.2 Connected Components and Patch Merging

Spatially contiguous regions of the same landscape type were identified using connected-component labelling (`skimage.measure.label`). To avoid an excessive number of tiny fragments, patches smaller than 100 pixels ($\sim$0.09 km^2) were merged into their most similar neighbour using a mapping-table approach. This procedure yielded **166,464 landscape patches** covering the full study extent.

For each patch, the mean value of the 12 selected environmental features was computed from the underlying raster stack, producing a 166,464 $\times$ 12 node-feature matrix.

3.1.3 Patch Network

An undirected graph was constructed by connecting patches that share at least one boundary pixel. The resulting `NetworkX` graph contains **166,464 nodes** and **465,833 edges**. Node attributes are the mean environmental features; no edge weights were used.

3.2 Baseline: Random Forest

A per-species Random Forest (Breiman, 2001) classifier served as the non-spatial baseline. For each of the 3,756 qualifying species ($\geq$ 100 GBIF records), the following procedure was applied:

1. **Presence labels.** GBIF occurrence coordinates were mapped to the patch-label raster, yielding a set of presence patches per species.
2. **Pseudo-absence generation.** Background patches were sampled at a 3:1 ratio (absence : presence) from patches without recorded occurrences, using a fixed random seed for reproducibility.
3. **Model training.** A `scikit-learn` Random Forest was trained on the patch-level mean features using tuned hyperparameters (see below).
4. **Evaluation.** Performance was assessed via 5-fold cross-validated metrics: AUC, accuracy, precision, recall, F1, MCC, Cohen's κ, and TSS.

Hyperparameter tuning was performed via `GridSearchCV` on the same six representative species used for the GNN architecture search, covering diverse habitat types across Switzerland. The best hyperparameters from this limited tuning were then applied uniformly to all 3,756 species.

3.3 GNN-SDM

3.3.1 Model Architecture

The SOM-GNN-SDM is based on GraphSAGE (Hamilton et al., 2017), implemented in PyTorch Geometric (Fey and Lenssen, 2019). At each layer l, a node v aggregates information from its neighbourhood $\mathcal{N}(v)$ using the mean aggregator and combines it with its own representation:

$$\mathbf{h}_{\mathcal{N}(v)}^{(l)} = \frac{1}{|\mathcal{N}(v)|} \sum_{u \in \mathcal{N}(v)} \mathbf{h}_u^{(l-1)} \tag{1}$$

$$\mathbf{h}_v^{(l)} = \sigma\Big(\mathbf{W}^{(l)} \cdot \text{CONCAT}\Big(\mathbf{h}_v^{(l-1)},\ \mathbf{h}_{\mathcal{N}(v)}^{(l)}\Big)\Big) \tag{2}$$

where $\mathbf{h}_v^{(0)} = \mathbf{x}_v$ is the input feature vector of node v, $\mathbf{W}^{(l)}$ is a learnable weight matrix, and σ is the LeakyReLU activation function. Dropout ($p = 0.2$) is applied after each layer during training. The final node embeddings are passed through a linear layer with sigmoid activation to produce a habitat-suitability score $\hat{y}_v \in [0, 1]$ for every patch:

$$\hat{y}_v = \text{sigmoid}\Big(\mathbf{w}^\top \mathbf{h}_v^{(L)} + b\Big) \tag{3}$$

where L is the number of GraphSAGE layers.

An architecture search was conducted on six representative species spanning diverse habitats (*Picea abies*, *Fagus sylvatica*, *Trifolium pratense*, *Potentilla erecta*, *Phragmites australis*, *Senecio inaequidens*). Six architectures were compared:

- 1-layer: [16] (417 parameters)
- 2-layer: $[24, 12]$ (1,201 parameters)
- Paper default: $[24, 18, 8]$ (1,787 parameters)
- Wider: $[48, 32, 16]$ (5,361 parameters)
- Extra-wide: $[64, 48, 32]$ (10,929 parameters)
- 4-layer: $[32, 24, 16, 8]$ (3,417 parameters)

The extra-wide 3-layer architecture $[64, 48, 32]$ consistently achieved the highest mean AUC across all test species and was selected for production training. This is notably wider than the $[24, 18, 8]$ architecture proposed by Wu et al. (2025) for virtual fauna species, suggesting that flora—with their sessile nature and finer-grained habitat associations—benefit from greater representational capacity per layer rather than additional depth.

3.3.2 Training Procedure

For each species, the GNN-SDM was trained independently using the following pipeline:

1. **Background selection via random pseudo-absence.** Non-presence patches were sampled at a 3:1 ratio (background : presence) using a fixed random seed. An initial design based on One-Class SVM background selection (as in Wu et al., 2025) was evaluated during development but abandoned in favour of random pseudo-absence for the production run. The random strategy ensures a fair comparison with the RF baseline (which uses same pseudo-absence protocol) and proved more robust across the full species set, avoiding the computational overhead and occasional instability of fitting a kernel SVM on high-dimensional patch features for each of 3,756 species.

2. **PageRank weighting.** Presence patches were weighted by their PageRank centrality in the landscape graph, following Wu et al. (2025). This assigns higher importance to structurally well-connected patches, reflecting their greater ecological significance as habitat corridors or core areas. Background patches received uniform weight of 1.0.

3. **Train/validation split.** The labelled patches (presence + background) were shuffled and split 80/20 into training and validation sets.

4. **Optimisation.** The model was trained with the Adam optimiser (learning rate = 0.001) using a weighted mean-squared-error loss:
$$\mathcal{L} = \frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} w_i \, (\hat{y}_i - y_i)^2$$
where $\mathcal{T}$ is the training set, w_i the PageRank-derived weight, $\hat{y}_i$ the predicted suitability, and $y_i \in \{0, 1\}$ the label.

5. **Early stopping.** Validation AUC was evaluated every 10 epochs. Training was halted if no improvement occurred for 50 consecutive epochs (patience). The model state with the best validation AUC was restored for prediction. Maximum training duration was 500 epochs.

6. **Prediction.** The best model was used to predict habitat suitability for all 166,464 patches, yielding a continuous suitability surface per species.

3.3.3 Evaluation

To evaluate the GNN-SDM on the same footing as the RF baseline, a post-hoc evaluation was performed for each species: presence patches and a 3:1 random sample of non-presence patches were combined, and the predicted suitability scores were evaluated using the same eight metrics (AUC, accuracy, precision, recall, F1, MCC, Cohen's κ, TSS) with a classification threshold of 0.5.

3.3.4 Computational Setup

Production training of all 3,756 species was executed as an AWS SageMaker Training Job on a GPU instance (NVIDIA GPU, CUDA-accelerated). The job supported checkpointing for spot-instance interruptions and completed in approximately 50 minutes of GPU time. Node features were standardised (zero mean, unit variance) before training, and the full graph (166,464 nodes, 931,666 directed edges) was loaded onto the GPU for efficient message passing.

3.4 Conservation Prioritisation

The final stage of the pipeline combines the GNN-SDM habitat suitability predictions with circuit-theory connectivity analysis to produce a multi-species conservation priority surface, which is then overlaid with existing protected areas to identify protection gaps.

3.4.1 Complementarity of Suitability and Connectivity

A natural question arises as to whether the circuit-theory step is redundant given that GraphSAGE already incorporates neighbourhood information during message passing. The two components are, however, fundamentally complementary:

- **GNN suitability** is a *local* property. GraphSAGE aggregates features from the 1–2 hop neighbourhood to predict whether a given patch is suitable habitat for a species. The output answers: "Is this place good habitat?"
- **Circuit-theory current flow** is a *global network* property. It treats the entire landscape graph as an electrical circuit and computes which patches act as irreplaceable bridges between isolated population clusters. The output answers: "Is this place critical for maintaining connectivity between populations?"

These two perspectives are orthogonal. A patch may have moderate suitability (GNN score $\sim$0.3) yet carry very high current because it lies in the only traversable corridor between two populations. Conversely, a patch in the core of a large population may have high suitability but low current because alternative paths exist. Conservation action is most impactful at patches that score highly on *both* dimensions—places that are simultaneously good habitat and vital connectivity links.

3.4.2 Circuit-Theory Connectivity

For each species, the landscape graph is treated as a resistor network where the conductance of each edge (u, v) is defined as the mean suitability of the two adjacent patches:

$$g_{uv} = \max\left(10^{-4},\ \frac{\hat{y}_u + \hat{y}_v}{2}\right) \tag{4}$$

This ensures that current preferentially flows through high-suitability regions (low resistance), reflecting the ecological principle that organisms move more easily through suitable habitat.

Source and ground nodes are defined by the species' actual observation patches. These patches are grouped into connected components in the landscape graph; components with ≥ 3 patches are ranked by size, and current flow is solved between the top $k = 6$ component pairs using Kirchhoff's circuit laws. Specifically, for each source–ground pair, one unit of current is injected uniformly across source nodes and extracted uniformly at ground nodes. The resulting system of linear equations (conductance Laplacian) is solved for node voltages using a sparse direct solver (`scipy.sparse.linalg.spsolve`), and the absolute current through each node is accumulated across all pairs.

Species with fewer than 2 qualifying components are excluded, as connectivity cannot be meaningfully assessed.

3.4.3 Species Selection and Filtering

The gap analysis targets threatened species (IUCN categories NT, VU, EN, CR) that have both GNN suitability scores and sufficient spatial structure for circuit analysis. Two filters are applied:

1. **Non-native exclusion.** Species whose IUCN threat status refers to their native range outside Switzerland (e.g. *Ginkgo biloba*, *Sequoiadendron giganteum*) are excluded, as their occurrences in Switzerland represent planted individuals in parks and gardens rather than populations requiring habitat connectivity.

2. **Minimum connectivity.** Species whose observation patches form fewer than 2 connected components of size ≥ 3 are excluded, as the circuit-theory framework requires at least one source–ground pair.

3.4.4 Normalisation and Aggregation

Per-species current flow values are normalised to $[0, 1]$ using the 99th percentile as ceiling:

$$\tilde{c}_v = \min\left(1,\ \frac{c_v}{P_{99}(c)}\right) \tag{5}$$

This percentile-based approach is more robust than max-normalisation, which is sensitive to single extreme pinch-point patches that would compress all other values toward zero.

Species are weighted by IUCN threat level (CR = 4, EN = 3, VU = 2, NT = 1) to reflect conservation urgency. The multi-species priority surface is then computed as:

$$P_v = \frac{1}{2}\left(\frac{\sum_s w_s \hat{y}_{v,s}}{\sum_s w_s} + \frac{\sum_s w_s \tilde{c}_{v,s}}{\sum_s w_s}\right) \tag{6}$$

where w_s is the IUCN weight of species s, $\hat{y}_{v,s}$ the GNN suitability, and $\tilde{c}_{v,s}$ the normalised current flow at patch v.

3.4.5 Protected Area Overlay

The Swiss protected area network (WDPA; 10,522 polygons) is rasterised onto the 30 m patch grid. Each patch is assigned the fraction of its constituent pixels that fall within any protected area polygon. The analysis is restricted to patches within Swiss territory (majority of pixels inside the national boundary) to avoid false negatives from neighbouring countries.

Conservation gaps are quantified by examining what fraction of high-priority patches (top 5% and top 10% of the priority surface) currently lack any protected-area coverage. These unprotected high-priority patches represent the most urgent candidates for protection expansion.

4 Results and Discussion

4.1 Landscape Graph Construction

The SOM converged after approximately 150 training epochs on the 400 000-pixel subsample (80 % of 500 000, with 20 % held out for validation). Figure 1 shows the evolution of validation quantisation error (QE) and topographic error (TE) across training stages. QE decreases steeply during the first 60 epochs and plateaus beyond epoch 150, confirming that additional training yields diminishing returns. Inspection of the codebook feature maps (Figure 5) corroborates this: at 150 epochs, the codebook displays smooth gradients with distinct regional structure, whereas at 200 epochs the features fragment into noise as the shrinking neighbourhood radius causes individual neurons to overfit local samples. While QE continues to decrease beyond 150 epochs, the topographic error increases and the codebook loses the spatial coherence needed for meaningful landscape-type clustering. The final validation metrics (QE = 1.54, TE = 0.14) at 150 epochs thus represent the best trade-off between quantisation fidelity and topological preservation.

PLACEHOLDER
som_convergence

Figure 1: SOM convergence on the held-out validation set. Left axis: quantisation error (mean L_2 distance to best-matching unit). Right axis: topographic error (fraction of samples whose two closest BMUs are non-adjacent on the grid). The dashed vertical line marks the 150-epoch production threshold.

Hierarchical clustering of the 4 096 codebook vectors (Ward's linkage, cut at 30 clusters) partitioned the SOM grid into 30 landscape types representing recurring environmental configurations across Switzerland. Connected-component labelling and merging of fragments smaller than 100 pixels produced the final landscape graph of **166 464 patches** and **465 833 edges**.

The patch size distribution (Figure 4) shows a median of approximately **X** pixels ($\sim\mathbf{X}\,\mathrm{km}^2$), with a long tail towards larger contiguous patches in homogeneous terrain (e.g. alpine grassland, dense forest). The minimum enforced size of 100 pixels ($\sim 0.09\,\mathrm{km}^2$) ensures that each patch is large enough to carry ecologically meaningful feature averages.

The full-extent PCA-coloured patch map (Figure 3) reveals coherent biogeographic structure: similar colours emerge for the Swiss Plateau and the Po Valley south of the Alps, reflecting analogous lowland environmental profiles. High-alpine zones form a distinct cluster, while the southern Pre-Alps and Ticino group with the Jura arc, both of which are characterised by their southward slope and intermediate elevations.

Figure 2 zooms into a ∼16 × 16 km region north of Bern. Urban patches (Bern city centre) are clearly delineated. The Dählhölzli forest along the Aare, the Schärmenwald near Ittigen, and the Tiefenau peninsula emerge as distinct forest-type patches. Overall, when compared with local knowledge, clear ecological structures are visible at this resolution.

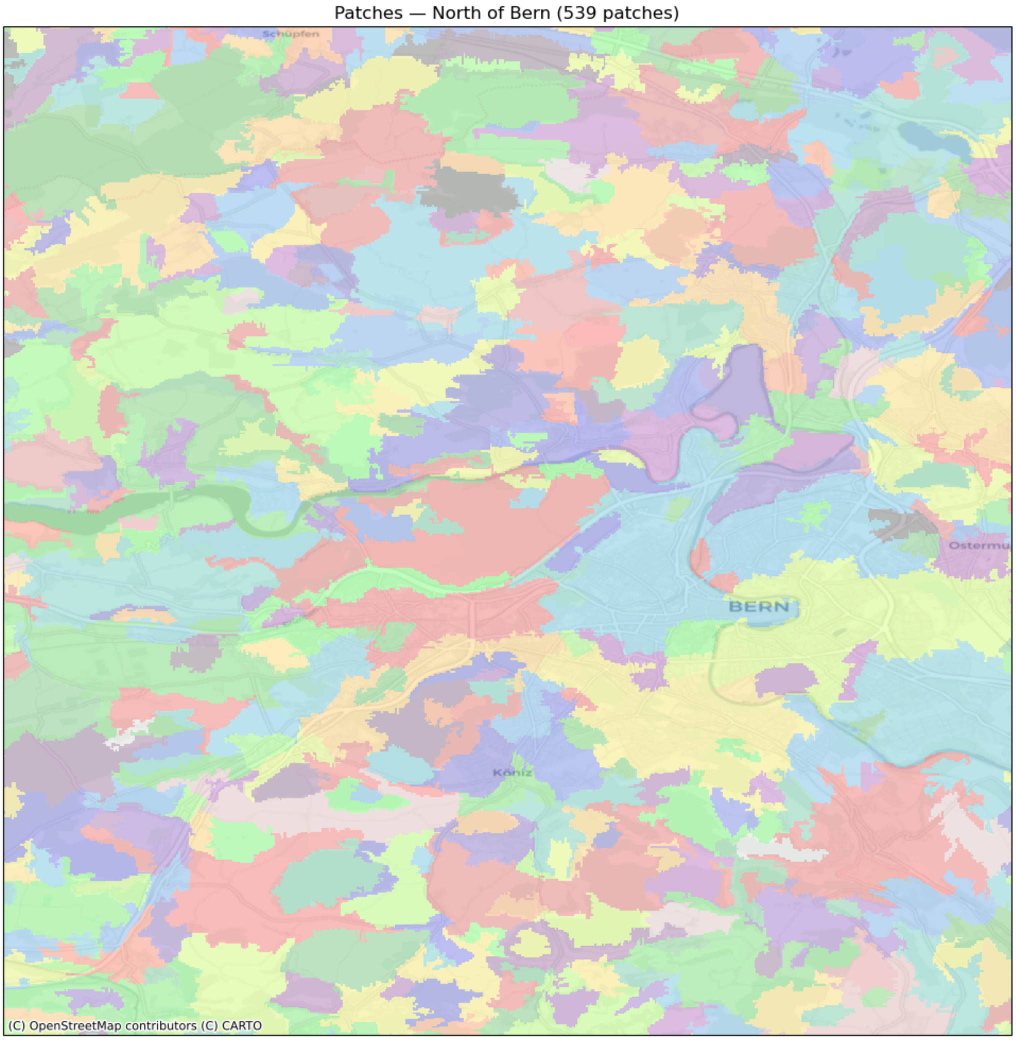

Figure 2: Landscape-type map for the Bern region. Colours represent the 30 landscape types derived from hierarchical clustering of the SOM codebook. The tessellation captures transitions between forested hillsides, agricultural plains, and built-up areas at 30 m resolution.

Discussion. The SOM-based discretisation serves a dual purpose: it reduces the ∼160 million valid pixels to a tractable graph of 166 464 nodes while ensuring that patch boundaries follow ecological gradients rather than an arbitrary regular grid. This adaptive resolution means that patches are smaller in heterogeneous landscapes (e.g. valley floors with interspersed land cover) and larger in homogeneous terrain (e.g. high-alpine meadows), mirroring the spatial structure relevant to species distributions.

A limitation of the adjacency-based edge construction is that connections between patches sharing only a single boundary pixel receive the same weight as connections along extensive shared borders. The graph therefore contains "bridge" edges representing tenuous spatial contact (e.g. two forest patches touching through a one-pixel corridor under a road). This is revisited in Section 5.1.

4.2 Random Forest Baseline

A per-species Random Forest served as the non-spatial baseline. Across 3 751 species, the RF achieved a mean AUC of 0.838 (median = 0.834), with 99.4 % of species exceeding 0.7. Full metric summaries are provided in Appendix B. The RF's key limitation is its independence assumption: each patch is classified without regard to its landscape context, so the model cannot distinguish isolated habitat fragments from patches embedded in connected corridors.

4.3 SOM-GNN-SDM Performance

TODO: GNN results and comparison with RF.

4.4 Architecture Search

TODO: Architecture discussion.

4.5 Common vs. Vulnerable Species

TODO: Species-level analysis.

4.6 Habitat Suitability Maps

TODO: Map figures.

4.7 Conservation Prioritisation

TODO: Conservation application results.

5 Limitations and Future Work

5.1 Limitations

TODO: Limitations.

5.2 Future Directions

TODO: Future directions.

6 Conclusion

TODO: Write conclusion.

Acknowledgements

TODO: Acknowledgements.

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A Supplementary Figures

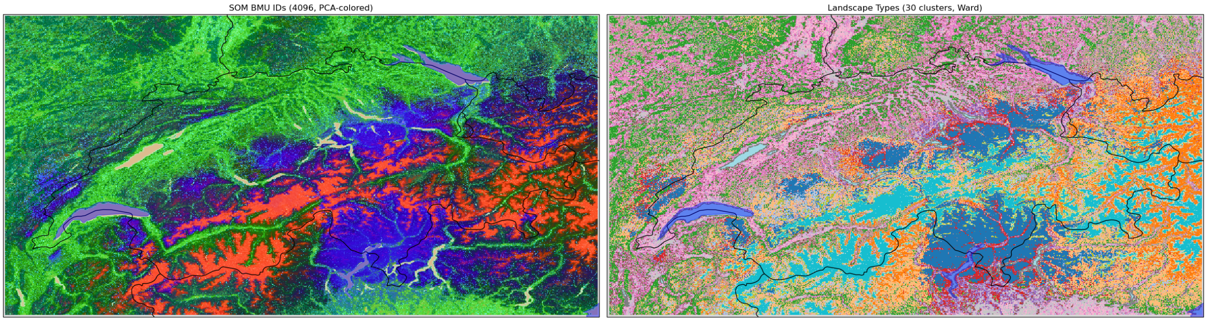

Figure 3: PCA-coloured landscape patch map of Switzerland. Codebook vectors are projected onto 3 principal components (mapped to RGB), so perceptually similar colours indicate environmentally similar patches. Major biogeographic zones are clearly distinguishable: lowland plateaus (Swiss Plateau, Po Valley) share warm tones, high-alpine areas appear as a distinct cool cluster, and forested pre-alpine slopes form intermediate gradients.

PLACEHOLDER
patch_size_distribution

Figure 4: Patch size distribution (log scale) for the 166 464 landscape patches. The vertical dashed line marks the minimum threshold of 100 pixels ($\sim$0.09 km^2).

B Random Forest Baseline Details

Table 2 summarises the cross-validated performance metrics for the Random Forest baseline across all 3 751 species.

Table 2: Random Forest baseline: summary statistics across 3 751 species (5-fold cross-validation, threshold = 0.5).

Metric	Mean	Median	Std
AUC	0.838	0.834	0.061
Accuracy	0.812	0.804	0.038
Precision	0.715	0.713	0.110
Recall	0.395	0.369	0.175
F1	0.489	0.482	0.161
MCC	0.428	0.412	0.148
TSS	0.346	0.318	0.165

The AUC distribution is strongly right-concentrated: 99.4 % of species achieve AUC > 0.7, and 69.5 % exceed 0.8. The few species below 0.7 are extreme rarities with fewer than 10 presence patches, where cross-validation folds contain too few positive samples for stable evaluation.

Threshold-dependent metrics tell a different story. Recall averages only 0.395 and 71.7 % of species have recall below 0.5 at the default threshold of 0.5. This reflects the strong class imbalance: mean prevalence is ∼0.6 % (980 presence patches out of 166 464), so the classifier defaults to predicting absence for borderline patches. AUC—which is threshold-independent—is therefore the more reliable performance indicator in this setting.

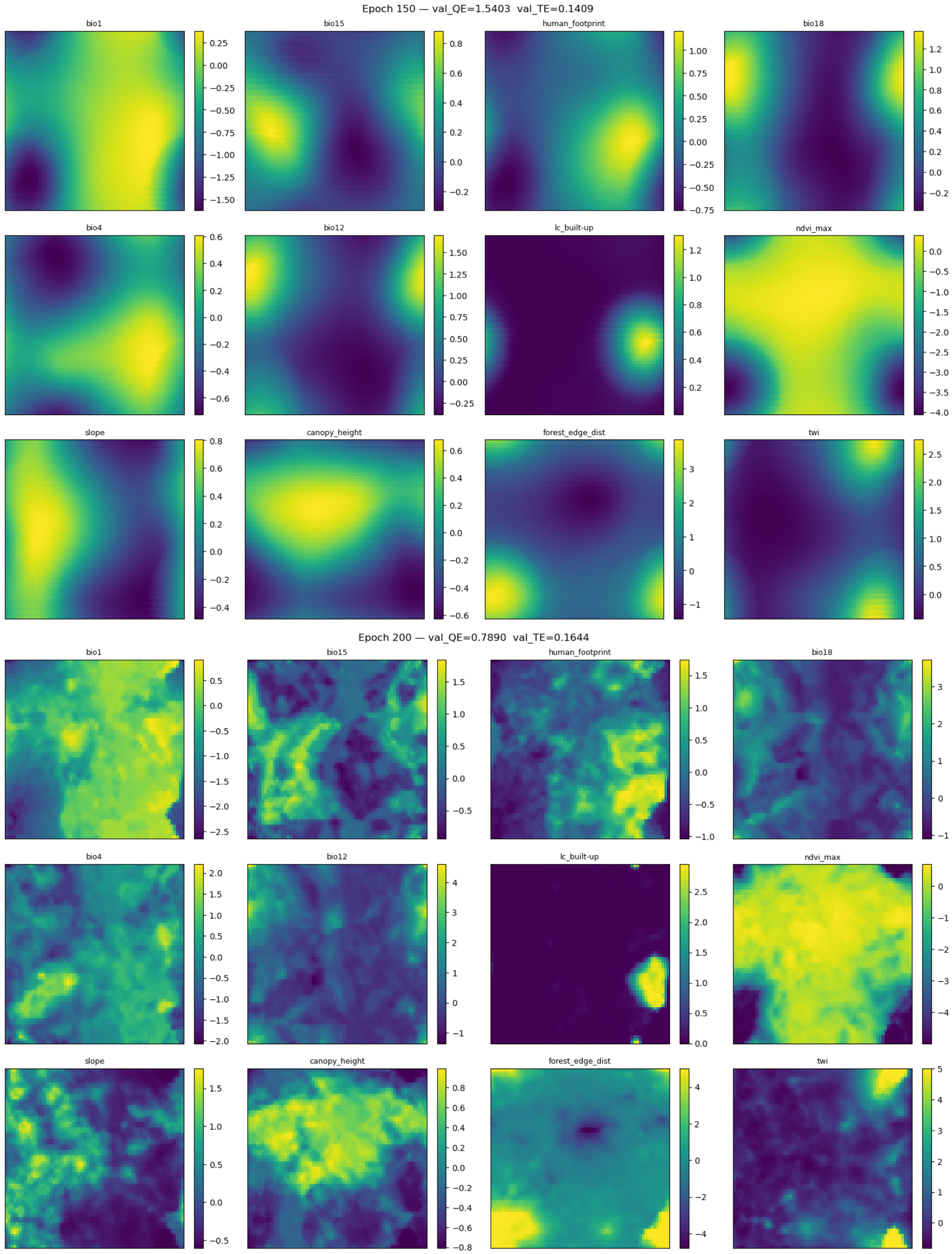

Figure 5: SOM codebook feature maps at epoch 150 (top) and epoch 200 (bottom). At 150 epochs the codebook shows smooth gradients with distinct regional structure across all 12 features. At 200 epochs, overtraining causes feature maps to fragment into noise—particularly visible in slope, TWI, and bio15—as the reduced neighbourhood radius allows individual neurons to chase local samples rather than preserving global topology.